Causal signal processing

In a specific signal processing, the relationship between the input signal $x \in \mathbf{R}^n$ and the measured b is defined by:

$$a_{11}x_1 = b_1, \quad a_{ii}x_i + \sum_{j=1}^{i-1} a_{ij}x_j = b_i, \quad i = 2, \ldots, n. \tag{1}$$

The system is called **causal** because we can view x_i as a function of x_j with $j = 1, \ldots, i-1$. If the index i represents time, it would mean that the signal at a current time is a function of others in the past.

For an $n \times n$ system, the coefficients a_{ij} are generated by a specific formula: $a_{ij} = \frac{1}{i+j}$ for $1 \leq i, j \leq n$. The relationship can be formulated as a linear system $Ax = b$. Write a function `solve_causal_system(n,b)` that solves (1).

Input Format

- $10 \leq n \leq 5000$ (positive integer)
- $b \in \mathbf{R}^n$: The constant vector in the equation, generated randomly.

Output Format

- $x \in \mathbf{R}^n$: The solution to $Ax = b$.
- $A \in \mathbf{R}^{n \times n}$: The coefficient matrix in $Ax = b$.

Implementation

Goal: Fill in the following function:

```
import ###

def solve_causal_system(n, b):

  return x, A
```

Make sure you import any library you use to solve the problem.

Hint

- Observe the structure of A.

Warnings

- You should avoid using **np.linalg.inv()** to solve a system of equations because it less accurate and computationally expensive.